ClipSAT : Contest Math & Olympiad AMC 10/12, Kangaroo & Mawhiba

Unit 1.01: Olympiad Algebraic Manipulation, Vieta & Inequalities Name: _____

Q1. The coordinate diagram shows the parabolic curve f_1x = x^2 - 6x + 20 in the domain of Olympiad Algebra, Vieta & Inequalities. By completing the square f_1x = x - h^2 + k , find the sum k + f_15 .

[ ] A: 26
[ ] B: 29
[ ] C: 15
[ ] D: 24

Q2. A mathematical model in Olympiad Algebra, Vieta & Inequalities is defined by the cubic function y_2x = 2x^3 - 3x^2 + 15x - 8 . Calculate the exact gradient of the tangent to the curve at the point where x = 3 .

[ ] A: 69
[ ] B: 51
[ ] C: 36
[ ] D: 27

Q3. In Olympiad Algebra, Vieta & Inequalities, two displacement vectors are given by mathbf{u}_{3} = langle 3, -8 rangle and mathbf{v}_{3} = langle 9, 7 rangle . Compute the exact scalar value of mathbf{u}_{3} cdot mathbf{v}_{3} + |mathbf{u}_{3}|^2 .

[ ] A: -29
[ ] B: 73
[ ] C: 44
[ ] D: 156

Q4. The linear growth profile shown in the diagram for Olympiad Algebra, Vieta & Inequalities follows an arithmetic progression with first term a_1 = 4 and common difference d = 6 . Find the exact sum S_{15} of the first 15 terms.

[ ] A: 88
[ ] B: 780
[ ] C: 1320
[ ] D: 690

Q5. In Olympiad Algebra, Vieta & Inequalities, evaluate the exact definite integral I_5 = int_0^{2} 18x^2 + 12x,mathrm{d}x .

[ ] A: 72
[ ] B: 96
[ ] C: 84
[ ] D: 37

Q6. In Olympiad Algebra, Vieta & Inequalities, consider the quadratic function f_6x = x^2 - 8x + 99 . Express f_6x in the completed-square form x - h^2 + k and compute the exact value of k + f_611 .

[ ] A: 219
[ ] B: 215
[ ] C: 132
[ ] D: 208

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Q7. The graph illustrates the cubic model y_7x = 2x^3 - 4x^2 + 33x - 15 studied in Olympiad Algebra, Vieta & Inequalities. Find the instantaneous gradient frac{mathrm{d}y}{mathrm{d}x} of the tangent to the curve at x = 8 .

[ ] A: 417

[ ] B: 320

[ ] C: 353

[ ] D: 73

Q8. In Olympiad Algebra, Vieta & Inequalities, two displacement vectors are given by mathbf{u}_{8} = langle 4, -7 rangle and mathbf{v}_{8} = langle 8, 5 rangle . Compute the exact scalar value of mathbf{u}_{8} cdot mathbf{v}_{8} + |mathbf{u}_{8}|^2 .

[ ] A: -3

[ ] B: 65

[ ] C: 132

[ ] D: 62